Exercise 5

- Table 1

Q1 : SELECT employee_id,
name,
IFNULL(salary, 0) AS salary_with_default
FROM employees

employee_id	name	salary_with_default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

Q2 : SELECT employee_id,
name
IFNULL(department, 'Not Assigned') AS department_name
FROM employees

employee_id	name	department_name
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance

Table 2 -

Q3 : SELECT order_id,
customer_id
FROM orders
WHERE customer_id IS NULL;

Order_id	Customer_id
103	NULL

Q4 : SELECT order_id,
customer_id
IFNULL(delivery_date, 'Pending') AS delivery_status
FROM orders ;

order_id	customer_id	delivery_status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

Table 3

Q5: SELECT student_id,
name
IFNULL(grade, 0) AS final_grade
FROM students;

student_id	name	final_grade
1	Ethan	85
2	Maya	0
3	Olivia	90

Q6: SELECT COUNT(student_id) AS not_graded_count
FROM students
WHERE grade IS NULL;

not_graded_count
1

Table 4

Q7: SELECT product_id,
name,
price - IFNULL(discount, 0) AS final_price
FROM products;

product_id	name	final_price
S01	Keyboard	25
S02	Mouse	10
S03	Monitor	100

Table 5

Q8: SELECT count(*) AS missing_email_count
FROM customers
WHERE email IS NULL;

missing_email_count
2

Q9 : SELECT customer_id,
name,
IFNULL (email, 'No email') AS email_display
FROM customers;

customer_id	Name	email_display
1	Linda	No email
2	Joseph	joseph@mail.com
3	Mia	No email

Table 6

Q10 : SELECT payment_id,
IFNULL (method, 'Unknown') AS method_display,
Status
FROM payments;

payment_id	method_display	Status
301	Credit	NULL
302	PayPal	Success
303	Unknown	Failed

when counting NULL -> use COUNT(*) with WHERE column IS NULL.

Q11 : SELECT item_id,
item_name,
IFNULL (quantity, 0) AS quantity_checked
FROM inventory;

item_id	item_name	quantity_checked
1	Pen	0
2	notebook	150
3	Eraser	0

Q12 : SELECT emp_id,
COALESCE(bonus, commission) AS first_available_reward
FROM employee_extra;

emp_id	first_available_reward
1	300
2	200
3	NULL

Q13 : SELECT COUNT(*) AS no_room_count
FROM classes
WHERE room IS NULL;

no_room_count
2

Q14 : SELECT student_id,
date,
IFNULL(status, 'Not Marked') AS attendance_status
FROM attendance;

student_id	date	attendance_status
1	2025-04-01	Not Marked
2	2025-04-01	Present
3	2025-04-01	Absent.

Q15 : SELECT account_id,
IFNULL(account_type, 'Unknown') AS type_display,
IFNULL(balance, 0) AS balance_checked
FROM bank_accounts;

Account_id	type_display	balance_checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

Q16 : SELECT project_id,

title,

IFNULL(start_date, 'TBD') AS start_display

From projects ;

project_id	title	start_display
1	Website Revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

Q17 : SELECT review_id,

product_id,

IFNULL(comment, 'No Comment') AS comment_display,

IFNULL(rating, 0) AS rating_display

From reviews ;

review_id	product_id	comment display	rating display
1	301	great product	4
2	302	No Comment	0
3	303	Works fine	3

Q18 : SELECT supplier_id,

name,

COALESCE(phone, alt_phone, 'No Comment') AS contact_number

From suppliers ;

supplier_id	Name	contact_number
1	Global goods	123456789
2	Best Supplies	9876543210
3	Value Source	No Contact.

Q19 : SELECT user_id,

IFNULL(theme, 'Light') AS theme_set,

IFNULL(language, 'English') AS language_set,

IFNULL(timezone, 'UTC') AS timezone_set

From user_settings ;

user_id	Theme_set	language_set	timezone set
1	light	English	UTC
2	Dark	English	UTC+1
3	light	English	UTC

Q20 : SELECT record_id,

machine_id,

IFNULL(issue, 'Unknown Issue') AS issue_log,

IFNULL(technician, 'Not Assigned') AS technician_name

From maintenance ;

record_id	machine_id	issue_log	technician_name
1	M101	overheating	Not Assigned
2	M102	Unknown issue	Not Assigned
3	M103	Jammed	Alex